

Agentic AI Mastery

Session 1 • Practical Assignment

Your First Agentic Workflow: Setting Up & Using Claude Code

Type: Practical Assignment	Session: 1
Date Assigned: May 14, 2026	Time Estimate: Varies

Overview

This assignment gets you hands-on with Claude Code and Cursor, the tools you'll use to build agents throughout the cohort. You'll set up your development environment, connect Claude as your AI pair programmer, and complete your first agentic task using Agent mode. This is about building comfort with the tools and the goal→execution→review loop.

Learning Objectives

- Successfully install and configure Cursor with Claude API
- Understand the three Cursor modes (Chat, Autocomplete, Agent) and when to use each
- Execute a multi-step task using Agent mode
- Document your workflow and reflect on the agent behavior
- Identify opportunities and boundaries for agent autonomy in code tasks

Instructions

1. Watch Session 1 Video 3 (Claude Code Setup) completely before starting
2. Install Cursor from cursor.com for your operating system
3. Create an Anthropic API key at console.anthropic.com (free tier works)
4. Connect Claude in Cursor Settings → Models → Add Claude
5. Complete all three practice tasks below
6. Document your experience in a 300-word reflection
7. Submit screenshots, code samples, and reflection to the LMS by the deadline

Questions & Tasks

Q1: Task 1 (15-20 min): Open Cursor and switch to Chat mode. Ask Claude to explain how web scraping works, then ask it to identify security concerns. Document your three best questions and Claude's answers.

Q2: Task 2 (25-30 min): Give Agent mode this prompt: "Create a Python script that: - Reads data from a CSV file - Filters rows where a specific column meets a condition - Calculates summary statistics (mean, count, max) - Saves results to a new CSV file" Watch Claude reason through the steps, write the code, and offer to run it. Document: How did Agent mode break down the problem? What did it do well? What would you have done differently?

Q3: Task 3 (40-50 min): Use Agent mode to build a practical tool for YOUR work: Choose one: A) A script that processes data you actually use (CSV, JSON, API) B) A tool that automates a repetitive task you do weekly C) A formatter or analyzer for content you create Give Claude a clear goal and let Agent mode execute. Document: What autonomy level did you use (approve each step vs. let it run)? Where did you intervene? How would you deploy this?

Q4: Reflection: REFLECTION: YOUR FIRST AGENTIC WORKFLOW (300 words) Reflect on your experience with Agent mode across the three tasks: 1. THE GOAL→EXECUTION→REVIEW LOOP - Describe your workflow:... (see document)

Evaluation Rubric

Criterion	Excellent (4)	Good (3)	Developing (2)	Needs Work (1)
Environment Setup	Cursor installed, Claude AI connected, Agent mode setup	Tools installed, Agent mode setup	Tools installed, Agent mode setup	Tools not installed, Agent mode not setup
Task 1: Chat Mode	Thoughtful questions; clear questions and responses; shows understanding of the problem	Clear questions and responses; shows understanding of the problem	Basic questions and responses; shows understanding of the problem	Superficial questions and responses; shows little understanding of the problem
Task 2: Agent Mode	Clear documentation of Agent mode execution; shows understanding of the problem	Clear documentation of Agent mode execution; shows understanding of the problem	Clear documentation of Agent mode execution; shows understanding of the problem	Unclear documentation of Agent mode execution; shows little understanding of the problem
Task 3: Real Tool	Practical, working tool for a real-world task; shows understanding of the problem	Practical, working tool for a real-world task; shows understanding of the problem	Practical, working tool for a real-world task; shows understanding of the problem	Unpractical, non-working tool for a real-world task; shows little understanding of the problem
Reflection Essay	Insightful; clear vision for the future; shows understanding of the problem	Insightful; clear vision for the future; shows understanding of the problem	Insightful; clear vision for the future; shows understanding of the problem	Uninsightful; unclear vision for the future; shows little understanding of the problem

Resources & References

- Video 3: Claude Code Setup — Cursor & Agent Mode Configuration (Session 1)
- Cursor Documentation: <https://docs.cursor.com>
- Anthropic API Keys: <https://console.anthropic.com/api-keys>
- Claude Code CLI: <https://github.com/anthropics/claude-code>
- Troubleshooting: Session 1 FAQ in the LMS

- Office Hours: Live Q&A; on [Date/Time] for setup help

Submit your assignment through the Learning Management System by the deadline. Questions? Reach out to the instructors.